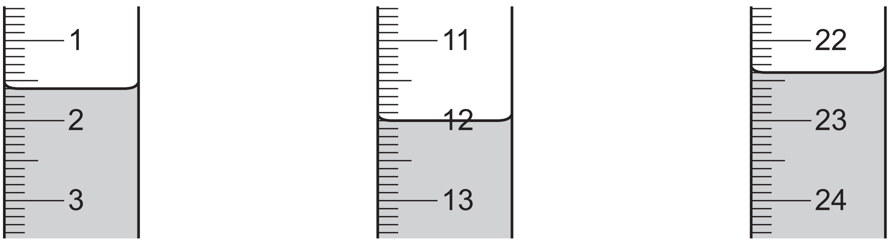
2 A student investigated the reaction between two different solutions of aqueous sodium carbonate, solution **K** and solution **L**, and dilute hydrochloric acid using two different indicators.

Two experiments were done.

Experiment 1

- A burette was rinsed with water and then with the dilute hydrochloric acid.
- The burette was filled with dilute hydrochloric acid. Some of the dilute hydrochloric acid was run out of the burette so that the level of the dilute hydrochloric acid was on the burette scale.
- Using a measuring cylinder, 25 cm^3 of solution **K** was poured into a conical flask.
- Five drops of methyl orange indicator **and** five drops of thymolphthalein indicator were added to the conical flask.
- The conical flask was placed on a white tile.
- Dilute hydrochloric acid was added slowly from the burette to the conical flask, while the flask was swirled, until the solution turned yellow. This is the first colour change.
- More dilute hydrochloric acid from the burette was added to the conical flask, while swirling the flask, until the solution changed colour again. This is the second colour change.

(a) Use the burette diagrams to complete the table for Experiment 1.



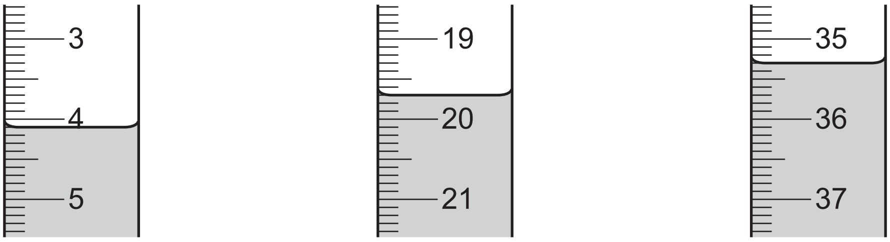
initial burette reading　　burette reading at first colour change　　burette reading at second colour change

	Experiment 1
burette reading at first colour change / cm^3	
final burette reading at second colour change / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid added for first colour change / cm^3	
total volume of dilute hydrochloric acid added for second colour change / cm^3	

[3]

(b) *Experiment 2*
- The conical flask was emptied and rinsed with distilled water.
- Experiment 1 was repeated using solution **L** instead of solution **K**.

Use the burette diagrams to complete the table for Experiment 2.



initial burette reading | burette reading at first colour change | burette reading at second colour change

	Experiment 2
burette reading at first colour change / cm^3	
final burette reading at second colour change / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid added for first colour change / cm^3	
total volume of dilute hydrochloric acid added for second colour change / cm^3	

[3]

(c) State the colour change observed at the end-point when dilute hydrochloric acid is added to methyl orange in an alkaline solution.

from .... to .... [1]

(d) For Experiment 1, compare the volume of dilute hydrochloric acid needed for the first colour change with the volume of dilute hydrochloric acid for the second colour change.

....

.... [2]

(e) Compare the concentration of solution **K** used in Experiment 1 to the concentration of solution **L** used in Experiment 2.
Explain your answer.

....

....

.... [3]

(f) **(i)** Deduce the volume of dilute hydrochloric acid needed for the second colour change when Experiment 2 is repeated using 50 cm^3 of solution **L**.

.... [2]

(ii) State why using 50 cm^3 of solution **L** would cause a problem.

....

.... [1]

(g) State the advantage of using a pipette instead of the measuring cylinder in these experiments.

.... [1]

(h) Explain why the conical flask was swirled as the dilute hydrochloric acid was added from the burette.

....

.... [1]

(i) At the start of Experiment 1, the burette was rinsed with water and then with dilute hydrochloric acid.
At the start of Experiment 2, the conical flask was rinsed with water but **not** with solution **L**.

(i) Explain why the conical flask was rinsed with water.

....

.... [1]

(ii) Explain why the conical flask was **not** rinsed with solution **L** in Experiment 2.

....

.... [1]

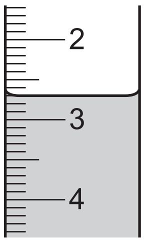
[Total: 19]

2 A student investigated the reaction between aqueous sodium hydroxide and two different solutions of dilute hydrochloric acid with different concentrations, labelled **Q** and **R**, using two different indicators.

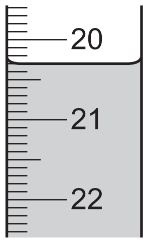
Three experiments were done.

(a) *Experiment 1*
- A burette was filled with dilute hydrochloric acid **Q**. Some of the dilute hydrochloric acid was run out of the burette so that the level of the dilute hydrochloric acid was on the burette scale.
- Using a measuring cylinder, 25 cm^3 of aqueous sodium hydroxide was poured into a conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- The conical flask was placed on a white tile.
- Dilute hydrochloric acid was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 1.



initial reading



final reading

	Experiment 1
final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid **Q** added / cm^3	

Experiment 2

- The conical flask was emptied and rinsed with distilled water.
- The burette was rinsed with distilled water and then with dilute hydrochloric acid **R**.
- Experiment 1 was repeated using dilute hydrochloric acid **R** instead of dilute hydrochloric acid **Q**.

Use the burette diagrams to complete the table for Experiment 2.

initial reading　　　　final reading

	Experiment 2
final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid **R** added / cm^3	

Experiment 3

- The conical flask was emptied and rinsed with distilled water.
- Experiment 2 was repeated using thymolphthalein indicator instead of methyl orange indicator.

Use the burette diagrams to complete the table for Experiment 3.

initial reading　　　　final reading

	Experiment 3
final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid **R** added / cm^3	

[5]

(b) Determine the simplest whole number ratio of the volumes of dilute hydrochloric acid **R** used in Experiment 2 and Experiment 3.

.... [1]

(c) Deduce the volume of dilute hydrochloric acid **Q** needed when Experiment 1 is repeated using thymolphthalein indicator instead of methyl orange indicator.

volume of hydrochloric acid **Q** = ... [2]

(d) Compare the concentration of dilute hydrochloric acid **Q** used in Experiment 1 to the concentration of dilute hydrochloric acid **R** used in Experiment 2.
Explain your answer.

....

....

....

.... [3]

(e) State how the results change, if at all, if the aqueous sodium hydroxide is warmed before adding the dilute hydrochloric acid.
Give a reason for your answer.

effect on results ....

reason ....
[2]

(f) State the advantage of using a pipette instead of the measuring cylinder in these experiments.

.... [1]

(g) Explain why a white tile is used in these experiments.

.... [1]

(h) At the start of Experiment 2 the burette was rinsed with distilled water and then with dilute hydrochloric acid **R**.

(i) State what was removed from the burette when it was rinsed with distilled water.

.... [1]

(ii) State what was removed from the burette when it was rinsed with dilute hydrochloric acid **R**.

....

.... [1]

(iii) Explain why the burette does **not** need to be rinsed at the start of Experiment 3.

.... [1]

(i) After the burette was filled with dilute hydrochloric acid at the start of Experiment 1, some of the acid was run out of the burette.

One reason for running the acid out of the burette is to make sure the level of the hydrochloric acid is on the scale.

Give one **other** reason why it is important to run some acid out of the burette after it has been filled for the first time in an experiment.

....

.... [1]

[Total: 19]

4 When solution **A** and solution **B** are mixed they react slowly to form iodine.
Starch solution is added to the mixture to act as an indicator.
When a certain amount of iodine is made there is a sudden colour change to blue-black.

Plan an investigation to find the effect of temperature on the rate of the reaction between solution **A** and solution **B**.

You are provided with solution **A**, solution **B**, starch solution and common laboratory apparatus.

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.... [6]

4 A student wanted to make some zinc chloride crystals.

The student followed the procedure shown.

step 1 Add excess zinc powder to dilute hydrochloric acid to form aqueous zinc chloride.

step 2 Remove unreacted zinc powder from the aqueous zinc chloride.

step 3 Heat the solution until it is saturated.

step 4 Allow the saturated solution to cool and remove the crystals that form.

(a) Write the equation for the reaction in **step 1**. Include state symbols.

.... [3]

(b) Explain why **excess** zinc powder is added in **step 1**.

.... [1]

(c) Suggest how unreacted zinc powder is removed in **step 2**.

.... [1]

(d) A saturated solution is formed in **step 3**.

Suggest what is meant by the term *saturated solution*.

....

.... [2]

(e) Explain why crystals form as the solution cools in **step 4**.

.... [1]

(f) Name **two** zinc compounds which react with dilute hydrochloric acid to form zinc chloride.

....

.... [2]

(g) If excess calcium metal is used instead of excess zinc powder in **step 1**, pure calcium chloride crystals do **not** form.

Explain why.

....

.... [1]

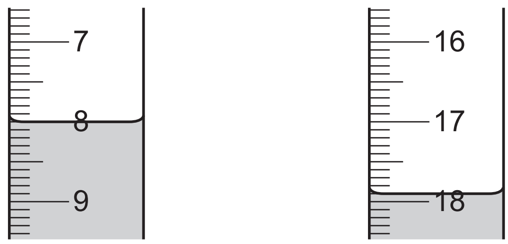
2 A student investigated the reaction between aqueous potassium hydroxide and two different aqueous solutions of hydrochloric acid labelled solution **A** and solution **B**.

Two experiments were done.

(a) *Experiment 1*

- A burette was filled with solution **A**. Some of solution **A** was run out of the burette so that the level of solution **A** was on the burette scale.
- A measuring cylinder was used to measure 25 cm^3 of the aqueous potassium hydroxide.
- The aqueous potassium hydroxide was poured into a conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- Solution **A** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 1.



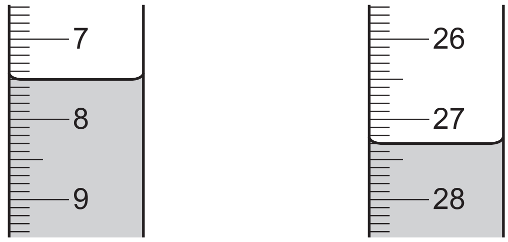
initial reading final reading

	Experiment 1
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **A** added / cm^3	

Experiment 2

- The conical flask was emptied and rinsed with distilled water.
- The burette was emptied and rinsed with distilled water.
- The burette was rinsed with solution **B**.
- The burette was filled with solution **B**. Some of solution **B** was run out of the burette so that the level of solution **B** was on the burette scale.
- A measuring cylinder was used to measure 25 cm^3 of the aqueous potassium hydroxide.
- The aqueous potassium hydroxide was poured into the conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- Solution **B** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 2.



initial reading final reading

	Experiment 2
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **B** added / cm^3	

[4]

(b) State the colour change observed in the conical flask at the end-point in Experiment 2.

from .... to .... [1]

(c) Before starting the titration in Experiment 2 the conical flask was rinsed with water.

(i) Explain why the conical flask was rinsed with water.

.... [1]

(ii) The conical flask was **not** then rinsed with aqueous potassium hydroxide.

State how rinsing the conical flask with aqueous potassium hydroxide would change the volume of solution **B** needed. Explain your answer.

....

.... [2]

(d) (i) Deduce which aqueous solution of hydrochloric acid, **A** or **B**, was more concentrated. Explain your answer.

....

.... [1]

(ii) Deduce how many times more concentrated this solution of hydrochloric acid was than the other solution of hydrochloric acid.

.... [1]

(e) Explain why Experiment 1 and Experiment 2 should be repeated.

....

.... [1]

(f) Deduce the volume of solution **B** required if Experiment 2 is carried out with 50 cm^3 of aqueous potassium hydroxide.

....

.... [2]

(g) Describe **one** change that could be made to the apparatus to improve the accuracy of the results.

....

.... [1]

(h) Describe what effect using a larger conical flask would have on the results obtained.

.... [1]

[Total: 15]

5 This question is about salts.

(a) Salts that are insoluble in water are made by precipitation.

- Lead(II) iodide, PbI_2, is insoluble in water.
- All nitrates are soluble in water.
- All sodium salts are soluble in water.

You are provided with solid lead(II) nitrate, $Pb(NO_3)_2$, and solid sodium iodide, NaI.

Describe how you would make a pure sample of lead(II) iodide by precipitation.

Your answer should include:
- practical details
- a chemical equation for the precipitation reaction.

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.... [5]

(b) Nitrates decompose when heated.

(i) When hydrated zinc nitrate is heated, oxygen gas is given off.

Describe a test for oxygen.

test ....

observations ....
[2]

(ii) Complete the equation for the decomposition of hydrated zinc nitrate.

$$2Zn(NO_3)_2{\bullet}6H_2O \rightarrow .....ZnO + ...NO_2 + O_2 + ...H_2O$$ [2]

5 This question is about salts.

(a) Salts that are soluble in water can be made by the reaction between insoluble carbonates and dilute acids. Zinc sulfate is soluble in water.

You are provided with solid zinc carbonate, $ZnCO_3$, and dilute sulfuric acid, H_2SO_4.

Describe how you would make a pure sample of zinc sulfate crystals.

Your answer should include:
- practical details
- how you would make sure that all the dilute sulfuric acid has reacted
- a chemical equation for the reaction.

....

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.... [5]

(b) Some sulfates decompose when heated.

When hydrated iron(II) sulfate is heated strongly, sulfur dioxide gas is given off.

(i) Describe a test for sulfur dioxide.

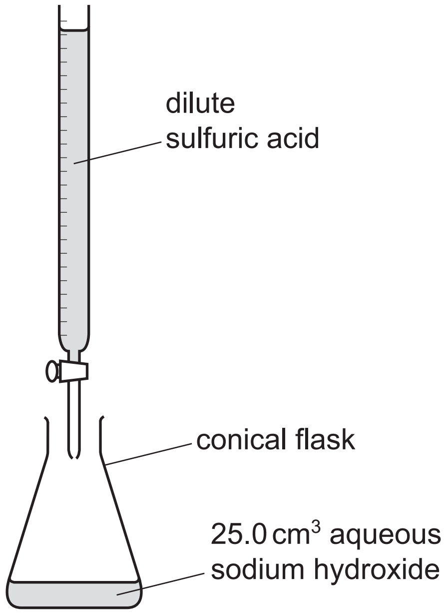
test ....

observations ....
[2]

(ii) Complete the equation for the decomposition of hydrated iron(II) sulfate.

$$.....FeSO_4{\bullet}7H_2O \rightarrow Fe_2O_3 + SO_2 + SO_3 + ...H_2O$$ [2]

5 (a) Dilute sulfuric acid and aqueous sodium hydroxide can be used to prepare sodium sulfate crystals using a method that involves titration.



(i) Suggest why universal indicator is **not** suitable for this titration.

.... [1]

(ii) Name an indicator that can be used in this titration.

.... [1]

20.0 cm^3 of dilute sulfuric acid neutralises 25.0 cm^3 of 1.00 mol / dm^3 aqueous sodium hydroxide. At the end of the titration the conical flask contains aqueous sodium sulfate with the dissolved indicator as an impurity.

(b) Describe how to prepare a **pure** sample of sodium sulfate crystals from the original solutions of dilute sulfuric acid and aqueous sodium hydroxide of the same concentrations.

You are not required to give details of how to carry out the titration.

....

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....

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....

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....

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.... [5]

(c) Sodium hydrogensulfate, $NaHSO_4$, dissolves in water to produce an aqueous solution, **X**, containing Na^+, H^+ and SO_4^{2-} ions.

State the observations when the following tests are done.

(i) A flame test is carried out on **X**.

.... [1]

(ii) Copper(II) oxide is warmed with an excess of **X**.

....

.... [2]

(iii) Acidified aqueous barium nitrate is added to **X**.

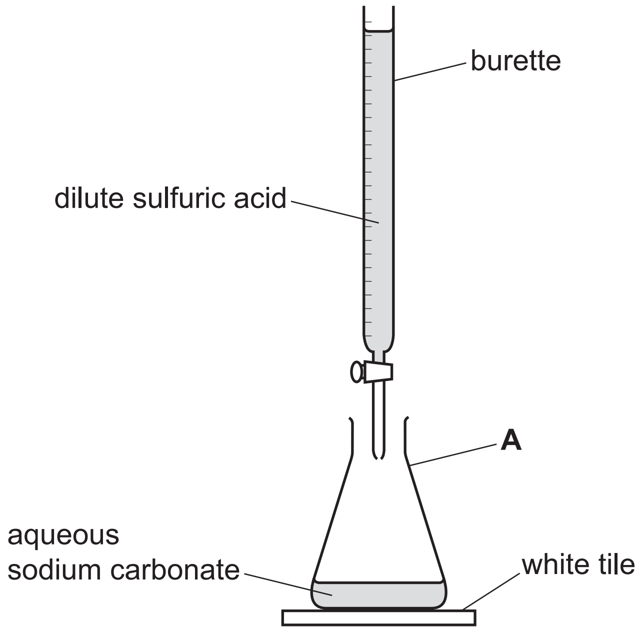
.... [1]

[Total: 11]

1 A student investigated the volume of dilute sulfuric acid that would react with 25.0 cm^3 of aqueous sodium carbonate.

- A burette was rinsed with water and then with dilute sulfuric acid.
- The burette was filled with dilute sulfuric acid. Some of the dilute sulfuric acid was run out of the burette so that the level of the dilute sulfuric acid was on the burette scale.
- 25.0 cm^3 of aqueous sodium carbonate was poured into the apparatus labelled **A** in the diagram.
- Five drops of methyl orange indicator were added to the aqueous sodium carbonate in **A**.
- The apparatus labelled **A** was placed on a white tile.
- The dilute sulfuric acid was added slowly to the 25.0 cm^3 of aqueous sodium carbonate until the colour of the methyl orange changed from yellow to orange.

The apparatus was arranged as shown in the diagram.



(a) Name the apparatus labelled **A**.

.... [1]

(b) State **one** safety precaution that should be taken when using dilute sulfuric acid.

.... [1]

(c) Give a reason why the white tile is used.

.... [1]

(d) Describe what should be done to the apparatus labelled **A** as the dilute sulfuric acid is added to the aqueous sodium carbonate.

.... [1]

(e) State why the burette was rinsed with water and then with dilute sulfuric acid at the start of the experiment.

water ....

....

dilute sulfuric acid ....

.... [2]

[Total: 6]

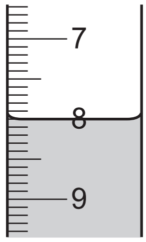
2 A student investigated the reaction between two different solutions of aqueous sodium carbonate, solution **K** and solution **L**, and two different solutions of dilute hydrochloric acid, acid **M** and acid **N**.

Three experiments were done.

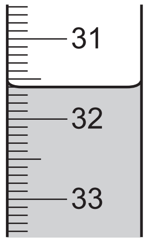
(a) *Experiment 1*

- A burette was filled with solution **K**. Some of solution **K** was run out of the burette so that the level of solution **K** was on the burette scale.
- Using a measuring cylinder 25 cm^3 of acid **M** was poured into a conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- The conical flask was placed on a white tile.
- Solution **K** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 1.



initial reading



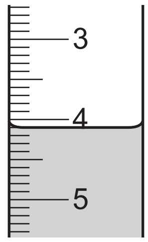
final reading

	Experiment 1
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **K** added / cm^3	

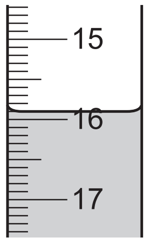
Experiment 2

- The conical flask was emptied and rinsed with distilled water.
- The burette was refilled with solution **K**. Some of solution **K** was run out of the burette so that the level of solution **K** was on the burette scale.
- Using a measuring cylinder 25 cm^3 of acid **N** was poured into the conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- The conical flask was placed on a white tile.
- Solution **K** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 2.



initial reading



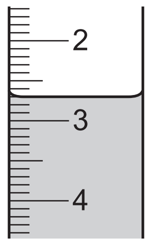
final reading

	Experiment 2
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **K** added / cm^3	

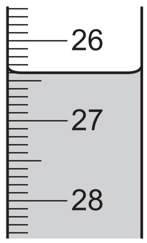
Experiment 3

- The burette was emptied and rinsed with distilled water.
- The conical flask was emptied and rinsed with distilled water.
- The burette was filled with solution **L**. Some of solution **L** was run out of the burette so that the level of solution **L** was on the burette scale.
- Using a measuring cylinder 25 cm^3 of acid **N** was poured into the conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- The conical flask was placed on a white tile.
- Solution **L** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 3.



initial reading



final reading

	Experiment 3
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **L** added / cm^3	

[5]

(b) State the colour change observed at the end-point in the conical flask in Experiment 1.

from .... to .... [1]

(c) Describe one **other** observation made when solution **K** was added to acid **M** in Experiment 1.

.... [1]

(d) **(i)** Compare the volumes of solution **K** used in Experiment 1 and Experiment 2.

....

.... [2]

(ii) Suggest why different volumes of solution **K** were needed in Experiment 1 and Experiment 2.

.... [1]

(e) Deduce the volume of solution **L** required to reach the end-point if Experiment 3 is repeated using acid **M** in place of acid **N**.

volume of solution **L** = ... cm^3 [1]

(f) Explain why the conical flask was rinsed with water at the start of Experiment 2 and Experiment 3.

.... [1]

(g) At the start of Experiment 3 the burette was rinsed with water.

Describe an additional step that should have been done after rinsing the burette with water but before filling the burette with solution **L**. Explain your answer.

....

.... [2]

(h) Explain why the conical flask is placed on a white tile.

.... [1]

(i) Describe how the reliability of the results can be confirmed.

....

.... [1]

(j) State **one** source of error in Experiment 1. Suggest an improvement to reduce this error.

source of error ....

improvement ....
[2]

[Total: 18]

4 This question is about reactions of bases and acids.

(a) Ammonia is a gas at room temperature.

What is the test for ammonia gas? Describe the positive result of this test.

test ....

result ....
[2]

(b) Ammonia reacts with water to form ions.

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

(i) How does this equation show that ammonia, NH_3, behaves as a base?

.... [1]

(ii) Aqueous ammonia is described as a weak base.

Suggest the pH of aqueous ammonia.

pH = ... [1]

(iii) Describe what is seen when aqueous ammonia is added to aqueous copper(II) sulfate, until no further change is seen.

....

....

.... [3]

(c) Aqueous sodium hydroxide, NaOH(aq), is a strong alkali that reacts with dilute sulfuric acid exothermically.

(i) What type of reaction is this?

.... [1]

(ii) Complete the equation for the reaction between aqueous sodium hydroxide and dilute sulfuric acid.

$$2NaOH + H_2SO_4 \rightarrow \text{.........................} + \text{.........................}$$

[2]

(d) A student wanted to find the concentration of some dilute sulfuric acid by titration. The student found that 25.0 cm^3 of 0.0400 mol / dm^3 NaOH(aq) reacted exactly with 20.0 cm^3 of H_2SO_4(aq).

(i) Name a suitable indicator to use in this titration.

.... [1]

(ii) Calculate the concentration of the H_2SO_4(aq) in mol / dm^3 using the following steps.

- Calculate the number of moles of NaOH in 25.0 cm^3.

moles = ...

- Deduce the number of moles of H_2SO_4 that reacted with the 25.0 cm^3 of NaOH(aq).

moles = ...

- Calculate the concentration of H_2SO_4(aq) in mol / dm^3.

concentration = ... mol / dm^3
[3]

(iii) Calculate the concentration of the 0.0400 mol / dm^3 NaOH(aq) in g / dm^3.

concentration = ... g / dm^3 [2]

[Total: 16]

(d) When copper is reacted with hot concentrated sulfuric acid, sulfur dioxide gas is formed.

Balance the chemical equation for this reaction.

$$Cu + ...H_2SO_4 \rightarrow CuSO_4 + SO_2 + ...H_2O$$ [1]

(e) Sulfur dioxide is a reducing agent.

Give the colour change that occurs when excess sulfur dioxide is bubbled into acidified aqueous potassium manganate(VII).

starting colour of the solution ....

final colour of the solution ....
[1]

(f) When sulfuric acid reacts with ammonia the salt produced is ammonium sulfate.

Write the chemical equation for this reaction.

.... [2]

(g) Barium sulfate is an insoluble salt.

Barium sulfate can be made from aqueous ammonium sulfate using a precipitation reaction.

(i) Name a solution that can be added to aqueous ammonium sulfate to produce a precipitate of barium sulfate.

.... [1]

(ii) Write an ionic equation for this precipitation reaction. Include state symbols.

.... [2]

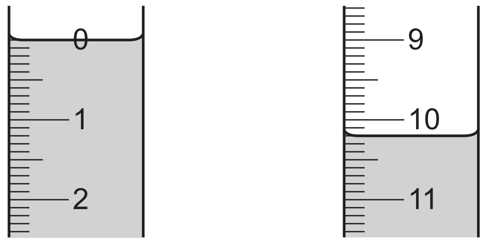
[Total: 16]

2 A student investigated the reaction between dilute hydrochloric acid and two different aqueous solutions of sodium carbonate, solution **E** and solution **F**.

Three experiments were done.

(a) *Experiment 1*
- A burette was filled up to the 0.0 cm^3 mark with dilute hydrochloric acid.
- Using a measuring cylinder, 25 cm^3 of solution **E** was poured into a conical flask.
- Five drops of thymolphthalein indicator were added to the conical flask.
- Dilute hydrochloric acid was slowly added from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 1.

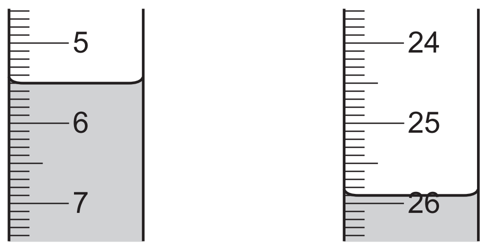


initial reading   final reading

final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid added / cm^3	

Experiment 2
- The conical flask was emptied and rinsed with distilled water.
- The burette was refilled with dilute hydrochloric acid.
- Experiment 1 was repeated using five drops of methyl orange indicator instead of thymolphthalein indicator.

Use the burette diagrams to complete the table for Experiment 2.



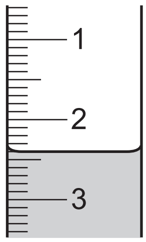
initial reading   final reading

final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid added / cm^3	

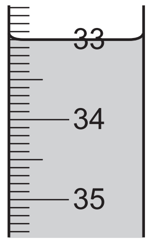
Experiment 3

- The conical flask was emptied and rinsed with distilled water.
- The burette was refilled with dilute hydrochloric acid.
- Using a measuring cylinder, 25 cm^3 of solution **F** was poured into the conical flask.
- Five drops of methyl orange indicator were added to the conical flask.
- Dilute hydrochloric acid was slowly added from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 3.



initial reading



final reading

final burette reading / cm^3	
initial burette reading / cm^3	
volume of dilute hydrochloric acid added / cm^3	

[5]

(b) What colour change was observed in the conical flask in Experiment 2?

from .... to ....

[2]

(c) Compare the volumes of dilute hydrochloric acid added in Experiment 2 and Experiment 3. Explain any difference.

....

.... [2]

(d) Determine the simplest whole number ratio of volumes of dilute hydrochloric acid used in Experiments 1 and 2.

ratio Experiment 1 : Experiment 2 = ... [1]

(e) What volume of dilute hydrochloric acid would be required if Experiment 3 was repeated using thymolphthalein indicator instead of methyl orange indicator?

volume = ... [2]

(f) The conical flask was rinsed with distilled water between each experiment.

(i) Why was the conical flask rinsed?

....

.... [1]

(ii) Why does it **not** matter if a little distilled water is left in the flask after it has been rinsed?

....

.... [1]

(g) State **two** sources of error in the experiments. For each error suggest an improvement that would reduce the error.

source of error 1 ....

improvement 1 ....

....

source of error 2 ....

improvement 2 ....

....
[4]

[Total: 18]

4 Many window-cleaning products contain aqueous ammonia. Aqueous ammonia is an alkali that reacts with dilute acids.

Plan an investigation to find which of two window-cleaning products contains the most concentrated aqueous ammonia. Include in your plan:

- the method you will use
- how your results will be used to determine which window-cleaning product contains the most concentrated aqueous ammonia.

You are provided with an aqueous solution of the two window-cleaning products, dilute hydrochloric acid of known concentration and common laboratory apparatus.

....

....

....

....

....

....

....

....

....

....

....

....

.... [6]

2 Soluble salts can be made by adding a metal carbonate to a dilute acid.

(a) Give the formula of the dilute acid which reacts with a metal carbonate to form a nitrate salt.

.... [1]

(b) A student wanted to make hydrated iron(II) sulfate crystals, $FeSO_4 \bullet xH_2O$, by adding excess iron(II) carbonate to dilute sulfuric acid. The student followed the procedure shown.

step 1 Add dilute sulfuric acid to a beaker.

step 2 Add small amounts of iron(II) carbonate to the dilute sulfuric acid in the beaker until the iron(II) carbonate is in excess.

step 3 Filter the mixture formed in **step 2**.

step 4 Heat the filtrate until it is a saturated solution. Allow to cool.

step 5 Once cold, pour away the remaining solution. Dry the crystals between filter papers.

(i) Why must the iron(II) carbonate be added in excess in **step 2**?

.... [1]

(ii) State **two** observations in **step 2** that would show that iron(II) carbonate was in excess.

1 ....

2 ....
[2]

(iii) Describe what should be done during **step 3** to ensure there is a maximum yield of crystals.

.... [1]

(iv) A saturated solution is formed in **step 4**.

Describe what a saturated solution is.

....

.... [2]

(v) Name a different compound that could be used instead of iron(II) carbonate to produce hydrated iron(II) sulfate crystals from dilute sulfuric acid.

.... [1]

(c) On analysing the crystals, the student found that one mole of the hydrated iron(II) sulfate crystals, $FeSO_4{\bullet}xH2O$, had a mass of 278 g.

Determine the value of *x* using the following steps:

- calculate the mass of one mole of $FeSO_4$

mass = ... g

- calculate the mass of H_2O present in one mole of $FeSO_4{\bullet}xH_2O$

mass of H_2O = ... g

- determine the value of *x*.

x = ...
[3]

(d) Insoluble salts can be made by mixing solutions of two soluble salts.

A student followed the procedure shown to make silver bromide, an insoluble salt.

step 1 Add aqueous silver nitrate to a beaker. Then add aqueous potassium bromide and stir.

step 2 Filter the mixture formed in **step 1**.

step 3 Dry the residue.

(i) State the term used to describe this method of making salts.

.... [1]

(ii) Give the observation the student would make during **step 1**.

.... [1]

(iii) Write the ionic equation for the reaction between aqueous silver nitrate and aqueous potassium bromide.

Include state symbols.

.... [3]

(e) Sodium chloride is an ionic salt. It can be made by reacting sodium with chlorine gas.

The equation for this reaction is shown.

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

Calculate the volume of chlorine gas, in cm^3, that reacts to form 2.34 g of NaCl.

The reaction takes place at room temperature and pressure.

volume of chlorine gas = ... cm^3 [3]

(f) Sodium chloride does not conduct electricity when solid, but does conduct electricity when molten.

(i) Explain why, in terms of structure and bonding.

....

....

....

.... [3]

(ii) Name the product formed at the positive electrode when electricity is passed through molten sodium chloride.

.... [1]

(iii) State the type of change that occurs at the positive electrode in **(ii)**.

Explain your answer in terms of electron transfer.

type of change ....

explanation ....
[2]

(iv) Describe what else can be done to sodium chloride to allow it to conduct electricity.

.... [1]

[Total: 26]

(c) Zinc reacts with dilute sulfuric acid to produce aqueous zinc sulfate.

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$

Hydrated zinc sulfate crystals are made from aqueous zinc sulfate.

Step 1 Solid zinc is added to dilute sulfuric acid until zinc is in excess.

Step 2 Excess zinc is separated from aqueous zinc sulfate by filtration.

Step 3 Aqueous zinc sulfate is heated until the solution is saturated.

Step 4 The saturated solution is allowed to cool and crystallise.

Step 5 The crystals are removed and dried.

(i) Name the residue in **step 2**.

.... [1]

(ii) In **step 3**, a saturated solution is produced.

Describe what a saturated solution is.

....

....

.... [2]

(iii) Name **two** compounds each of which react with dilute sulfuric acid to produce aqueous zinc sulfate.

1 ....

2 ....

[2]

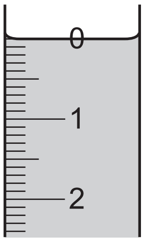
2 A student investigated the reaction between dilute ethanoic acid and two different solutions of sodium hydroxide labelled solution **A** and solution **B**.

Two experiments were done.

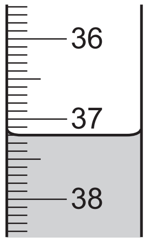
(a) *Experiment 1*

- A burette was rinsed with solution **A**.
- The burette was filled with solution **A**. Some of solution **A** was run out of the burette so that the level of solution **A** was on the burette scale.
- Using a measuring cylinder, 25 cm^3 of dilute ethanoic acid was poured into a conical flask.
- Five drops of thymolphthalein indicator were added to the conical flask.
- Solution **A** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 1.



initial reading



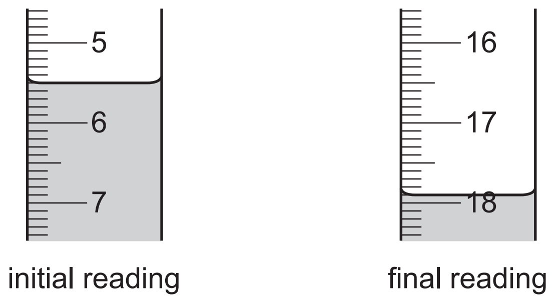
final reading

	Experiment 1
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **A** added / cm^3	

Experiment 2

- The conical flask was emptied and rinsed with distilled water.
- The burette was emptied and rinsed with distilled water.
- The burette was rinsed with solution **B**.
- The burette was filled with solution **B**. Some of solution **B** was run out of the burette so that the level of solution **B** was on the burette scale.
- Using a measuring cylinder, 25 cm^3 of dilute ethanoic acid was poured into a conical flask.
- Five drops of thymolphthalein indicator were added to the conical flask.
- Solution **B** was added slowly from the burette to the conical flask, while the flask was swirled, until the solution just changed colour.

Use the burette diagrams to complete the table for Experiment 2.



	Experiment 2
final burette reading / cm^3	
initial burette reading / cm^3	
volume of solution **B** added / cm^3	

[4]

(b) Explain why universal indicator is **not** a suitable indicator to use in this titration.

....

.... [1]

(c) **(i)** State which solution of sodium hydroxide, solution **A** or solution **B**, was the more concentrated.
Explain your answer.

....

.... [1]

(ii) State how many times more concentrated this solution of sodium hydroxide was than the other solution of sodium hydroxide.

....

.... [1]

(d) Determine the volume of solution **B** that would be required if Experiment 2 was repeated with 10 cm^3 of dilute ethanoic acid.

....

.... [2]

(e) Describe how the reliability of the results could be checked.

....

.... [1]

(f) A 25 cm^3 pipette can be used to measure the volume of a solution.

(i) Describe an advantage of using a 25 cm^3 pipette to measure the volume of the dilute ethanoic acid.

....

.... [1]

(ii) Explain why a 25 cm^3 pipette could **not** be used to measure the volume of solution **A**.

....

.... [1]

(g) **(i)** Explain why the burette was rinsed with distilled water in Experiment 2.

....

.... [1]

(ii) Explain why the burette was then rinsed with solution **B**.

....

.... [1]

(iii) State the effect that **not** rinsing the burette with solution **B** would have on the final burette reading.
Explain your answer.

effect ....

explanation ....

....
[2]

[Total: 16]